Ethan Yu

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Education

Queen's University Expected May 2029
BASc in Mechatronics and Robotics Engineering — Dean's Scholar *Kingston, Ontario*

Technical Skills

Programming Languages: C, C++, Python, MATLAB
Embedded Systems: ARM Cortex-M4, AVR, RTOS, Contiki OS, Raspberry Pi, SPI, UART, I2C
Hardware & Tools: Oscilloscope, Multimeter, Git, KiCad, LTspice, Onshape, SolidWorks

Experience

Undergraduate Research Assistant May 2026 – Aug. 2026
Queen's MEMS Laboratory *Kingston, Ontario*

- Developed embedded C firmware for a battery-powered wireless sensor node on the ATmega256RFR2 running Contiki OS, using **cooperative multitasking** for sensing, networking, and power management.
- Authored a SPI driver for a MAX31865 RTD-to-digital converter, including register-level configuration, conversion from ADC to temperature, and fault detection/clearing for a PT1000 temperature probe.
- Designed a request/response protocol over 6LoWPAN, RPL, and UDP that decodes compact codewords from a gateway to selectively collect and transmit only the requested sensor data.
- Implemented **deep-sleep power management** using AVR power-save mode and a wake timer to minimize energy use between transmissions, extending estimated battery life by over **17×**.

EngLinks Oct. 2025 – Present
STEM and Programming Tutor *Kingston, Ontario*

- Provided one-on-one tutoring across first-year and upper-year engineering courses to 15+ university students.
- Guided students through assignments and exam preparation across 3 university engineering courses.

Projects

Modbus Sensor Aug. 2026 – Sept. 2026

- Engineered embedded C firmware for a SAMG53-based sensor node that acquires and analyzes accelerometer data, transmitting processed measurements via **Modbus RTU** over RS485.
- Developed a SPI driver for a KX132-1211 3-axis accelerometer, using direct register-level PDC (DMA) control on the SPI peripheral to offload per-sample data transfer from the CPU.
- Implemented an RTC-driven acquisition scheduler collecting **9,000 samples** (3,000/axis) every 2 minutes, decoupling interrupt timing from blocking acquisition to prevent peripheral stalls.

Autonomous Rover Jan. 2026 – Apr. 2026

- Architected Arduino code for **closed-loop dual-motor speed control**, using encoder interrupts to estimate wheel velocities and PID controllers to regulate PWM and direction at 100 Hz.
- Programmed **Python ROS 2 nodes** on a Raspberry Pi to interface with Arduino over a custom 115200-baud serial protocol, sending wheel-velocity commands and receiving encoder counts for navigation.
- Implemented a 500 ms command timeout that resets wheel-speed targets and PID integrators when commands stop arriving, improving robustness to communication interruptions.

Passive Wireless Temperature Sensor Aug. 2026 – Sept. 2026

- Extended the open-source NanoVNA Saver software in Python with CSV export functionality, streamlining collection and analysis of frequency-response measurements from custom RLC PCBs.
- Designed and experimentally characterized multiple RLC PCB configurations with varying capacitance/inductance values to compare temperature-dependent changes in circuit bandwidth and Q factor.
- Calculated Q factor from measured resonant bandwidth across 24°C–160°C, identifying a configuration whose Q increased from 34.8 to 47.4 ($\Delta Q = 12.6$, approximately 36%).